Stand Up, Head Up, Stretch Up: Exploring Opportunities for Context-Aware Adaptive Stretching in MR for Microbreaks among Sedentary Workers

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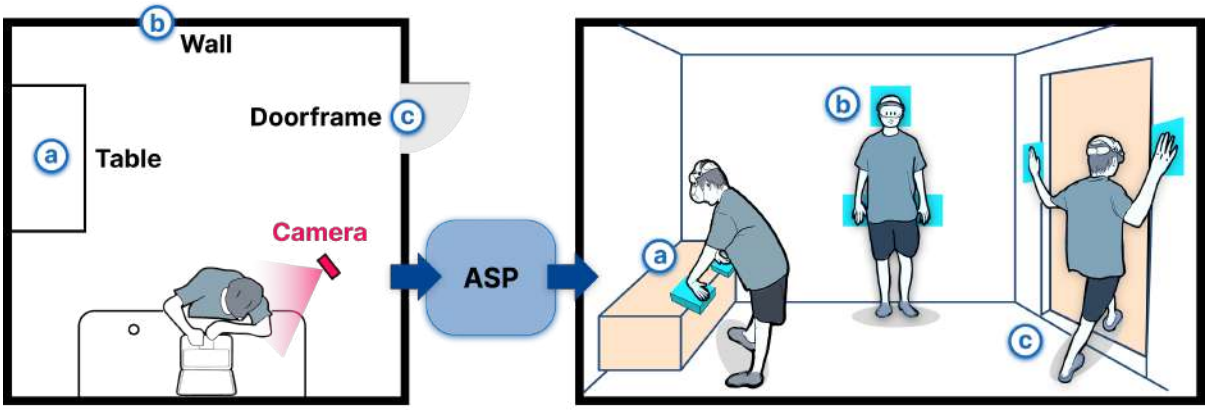


Fig. 1. System Overview. (Left) Our system captures the user's posture data using a camera. And it also captures the room feature that could support static stretching. (Right) It then generates interactive anchors through the ASP (Adaptive Stretching Pipeline) in the VR scene to guide stretching. The user is performing a table flexor wrist stretch, wall stand, and door frame chest stretch.

Prolonged desk work has become a major source of musculoskeletal strain, leading to fatigue, discomfort, and reduced mobility among sedentary professionals. Although prior interventions such as standing desks and exergames promote movement, most overlook cumulative postural imbalance and environmental constraints that shape ergonomic risk. We present Adaptive Stretching Pipeline (ASP), a mixed-reality (MR) system that delivers personalized stretching guidance during

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work breaks. ASP continuously monitors posture through joint-angle and duration analysis and leverages a vision-language model (VLM) to assess workspace supports such as desks or walls. Based on an integrated need-scoring model, the system generates adaptive MR stretching sessions targeting fatigued body regions during microbreak.
A three-phase formative study informed the system design, including expert interviews on clinical stretching practices, workplace observations to identify environmental supports, and stretch video analysis to construct a stretching library. A user study with 15 participants compared ASP to a Bend and randomised condition. Our results provide initial evidence that such adaptive mechanisms can shape perceived effort, engagement, and time perception in short-term interactions.

CCS Concepts: • **Human-centered computing** → *Interaction paradigms*; *Virtual reality*.

Additional Key Words and Phrases: Virtual/Augmented Reality, Empirical study that tells us about people, Lab Study, Input Technique

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1 INTRODUCTION

Laptops and desk-bound work have become central to modern productivity, but they come at a physical cost. Professionals now spend up to 75% of their workday seated at a computer [35, 56]. This extended sedentary behavior is strongly associated with musculoskeletal strain, including fatigue, reduced mobility, chronic discomfort, and elevated risk of injury [10, 20, 29]. Further, poor design of workstations, such as chairs that lack lumbar and neck support or desks set at improper heights, can result in extra ergonomic strain [1].

To mitigate such risks, researchers have introduced various ways to integrate movement into the workday, including under-desk ellipticals [14] and automated standing desks [23]. These interventions align with public health guidance linking physical activity to reduce chronic disease risk, reduce absenteeism, and improve workplace well-being [19]. The World Health Organization advocates for integrating movement into the workday, including flexible break schedules that support physical and mental recovery [59]. Moreover, the type of break also matters. Ding et al. found that standing and stretching breaks are more effective in reducing fatigue than passive or simple walking breaks [17]. Targeted interventions, such as neck-focused rehabilitation games, further underscore the benefits of goal-specific movement [44].

Prior research has introduced exergames designed to promote physical activity as a more engaging form of work break intervention [9]. Furthermore, some efforts have targeted specific body regions; for example, Mihajlovic et al. developed a serious game focused on rehabilitative neck movements [44]. Conventional screen-based exercise systems typically rely on non-immersive displays, where movements are guided through abstract visual instructions rather than situated bodily experience. Such setups provide limited support for embodied and spatially aware movement, making it difficult to align visual guidance with proprioceptive feedback and surrounding physical constraints [51, 61]. As a result, these systems are less well-suited for supporting safe and effective full-body stretching within short, self-directed breaks [7, 27].

In contrast, immersive XR technologies offer a fundamentally different interaction paradigm by tightly coupling visual perspective, body movement, and task feedback. Prior work has shown that immersion and perspective can significantly enhance players' sense of presence, embodiment, motivation, and performance, even when perceived exertion remains unchanged across conditions [7]. Similarly, Xu et al. reported that VR exergames lead to higher player performance compared to playing an identical game on a large display [61]. Both HMD-based and CAVE-based VR exergaming have been found to increase presence compared to non-VR exercise settings [51]. Immersive XR has been shown to motivate and engage users in physical activities. Ioannou et al. demonstrated that virtual augmentation of running and jumping enhances intrinsic motivation [27], while Born et al. found that players voluntarily sustained strenuous activities for longer durations when experiencing augmented exercise [8].

Beyond that, XR can sense and leverage the physical environment, integrating virtual guidance with real-world affordances such as walls, desks, or doorframes to support safe and feasible exercises [3, 31, 43]. These works also emphasize the importance of stationary supportive objects (e.g., walls, desks, or doorframes) for safe and effective stretching [31] and demonstrate that aligning physical and virtual affordances in MR exercise systems could shape effective and engaging experiences [3]. Recent toolkits like MRUK make it possible to embed interactive elements into real-world environments, opening up new opportunities for posture-aware and context-sensitive movement interventions [43].

While breaks and stretching are well established as beneficial for sedentary workers, and XR has demonstrated value for delivering embodied and immersive exercise experiences, current XR exercise systems primarily adapt based on in-system performance, such as task success or movement accuracy [53, 58]. These approaches do not account for users' accumulated ergonomic context, including posture persistence and strain that develop over the workday, nor do they systematically leverage real-world environmental supports that are critical for safe and effective stretching. These limitations highlight the need for XR systems that integrate postural sensing and environment-aware adaptation to deliver personalized, physically supportive stretching interventions.

In this paper, we present the Adaptive Stretching Pipeline (ASP), a mixed reality system that supports personalized stretching during workplace micro-breaks. ASP integrates posture sensing, environmental understanding, and adaptive action selection into a unified pipeline, enabling intervention based on both users' ergonomic state and spatial constraints. The system monitors user posture by analyzing joint angles and static durations, and also leverages a vision-language model (VLM) to assess environmental support factors, to estimate the need for stretching across different body regions. ASP also analyzes spatial features of the workspace to determine suitable areas for stretching. ASP then generates personalized 5-minute MR stretching sessions targeting fatigued body regions. To achieve this, we began with a three-phase formative study: (1) interviews with clinical practitioners to understand current practices related to muscular fatigue, diagnosis, and treatment, (2) observation of sedentary workers to identify and generate prompts for VLM of possible environmental supports in workplace settings, and 3) video analysis of the stretching video to determine the final set of stretching exercises used in this study. From this, we derived five design considerations that informed the development of ASP. To evaluate ASP, we conducted a user study with 15 participants, comparing our system to a commercial stretching application (Bend) and randomised condition. Results showed that ASP delivers a more tailored exercise experience, enhances engagement, and reduces time perception. We conclude by discussing findings of the study and offering design considerations for future adaptive, posture-aware break interventions.

2 RELATED WORK

2.1 Sedentary Work and Microbreak Interventions

In the context of occupational health, prolonged sitting has been consistently implicated in exacerbating lower back pain [46, 47]. There is a well-established link between the duration of computer use and an increased incidence of musculoskeletal symptoms [10, 29]. Over time, sustained postural demands can lead to muscular fatigue, reduced mobility, chronic discomfort, and elevated risk of injury [20]. Further, many people with bad backs gravitate towards sedentary work, poor design of workstations may then exacerbate their problems [1]. Also, ergonomic guidelines such as ISO 11226 [2] and RULA [42] suggest that when joint angles exceed specific thresholds and are maintained over time, they can induce significant ergonomic load.

Previous research describes how work breaks contribute to workplace wellbeing and productivity. Regular physical activity has been shown to reduce the risk of chronic conditions and, from an organizational standpoint, lower healthcare costs and employee absenteeism [19]. The World Health Organization recommends integrating more physical activity into the workday and allowing flexible, movement-friendly break schedules [59]. Regular breaks reduce accidents and physical discomfort in industrial settings and help prevent repetitive strain injury,

muscle fatigue, and prolonged sedentary behavior in office work [15, 55, 57]. Even short breaks have been shown to offer substantial benefits such as improving attention and reading comprehension [45]. Previous work has found that participants engaging in physical activities during work breaks [13]. Stretching is also a common physical exercise during short breaks, which has been shown to positively contribute to improved emotional state and muscular activation among office workers [24].

Despite these benefits, workers often forego taking breaks. To address these limitations, prior research has explored contextually-adaptive, just-in-time (JIT) interventions that deliver reminders or exercise prompts at opportune moments by accounting for users' activity patterns, attentional demands, and situational constraints [13]. These systems aim to reduce interruption cost and increase compliance by intervening when users are more cognitively and physically available.

However, prior work also suggests that even well-timed JIT interventions may suffer from low adherence when users lack intrinsic motivation or perceive the interventions as repetitive or unengaging [48]. This has motivated researchers to explore more engaging forms of work-break interventions, such as exergames, to complement timing-based adaptation. Prior research has introduced exergames designed to promote physical activity as a more engaging form of work break intervention [9]. Ren et al. proposed a posture-based interactive system called HealthSit, which was developed to promote a short lower-back stretching exercise during work breaks [50]. For example, Adolf et al. showed that a stretching system with real-time feedback helps improve the motivation of users [5]. Limber is an exergame that promotes movement, periodic stretching, and good posture among knowledge workers, but was overhauled to use a workstation-mounted Microsoft Kinect and ambient display because the hoodie clashed with the workplace dress code [49]. While these systems highlight the value of stretching, most prescribe fixed routines and fail to adapt to users' real-time ergonomic needs or environmental conditions. Immersive XR holds the potential to deliver more engaging and responsive microbreak experiences that dynamically adapt to the user and their surroundings.

2.2 Immersive and XR-Based Exercise Systems

An increasing number of studies have examined the exercise in XR, emphasizing the benefits of immersive experiences for physical activity [30]. Born et al. demonstrated that VR-based exergames provide greater motivation, embodiment, and performance compared to non-immersive versions [7]. Similarly, Xu et al. found that players performed better in VR exergames than when playing the same game on a large display [61]. These findings highlight the motivational and performance-enhancing potential of immersive XR systems. Several studies support this claim by showing how immersive augmentation influences player behavior. Ioannou et al. found that augmenting running and jumping in virtual environments enhances intrinsic motivation [28]. Dang et al. proposed using visual intervention to monitor and correct head posture during VR gameplay [16]. Likewise, Born et al. reported that users engaged in longer, more strenuous voluntary activities when virtual augmentation was present [8]. Stranick further showed that an adaptive VR-based stretching exergame designed for workers could improve both engagement and exercise outcomes [53].

While immersive XR offers motivational advantages, physical interaction plays a critical role in maintaining bodily awareness during exercise. Garcia Hernandez et al. argued that relying solely on virtual mid-air interactions may reduce proprioceptive feedback, as users' limbs are not in contact with tangible objects [21]. This suggests a need for incorporating physical support or props into XR-based exercise experiences. A recurring theme in these studies is the augmentation of existing home objects. Both Knierim et al. and Adiwangsa et al. noted users' preference for enhancing familiar household items with new interactive capabilities, suggesting that well-integrated objects can be repurposed to support exercise in the home [4, 32]. Knierim et al. explored how augmented reality (AR) can be seamlessly integrated into home environments, identifying use cases that align with people's routines and spatial constraints [32]. In line with this, Adiwangsa et al. emphasized that

home-based exercise systems should adapt to the evolving needs of individuals [4]. Furthermore, prior work has proposed design guidelines emphasizing the importance of aligning physical and virtual affordances in MR exercise systems. Ensuring that interactions feel natural and supported across both realms is crucial for effective and engaging experiences [3]. While XR systems enhance motivation and engagement, most adaptive XR exercise approaches focus only on adapting to in-system performance (e.g., task success and accuracy) rather than accumulated ergonomic context, such as posture persistence or fatigue. Moreover, they rarely leverage real-world environmental supports to improve safety and proprioceptive feedback.

2.3 Context-Aware and Adaptive Computing

Context-aware computing has long been a core theme in HCI and UbiComp research, aiming to enable systems that sense, interpret, and respond to user and environmental contexts. Prior studies have explored diverse modalities for human activity recognition, including camera-based, sensor-based, and gaze-based approaches [6, 33, 52]. Beyond implicit sensing, lightweight micro-interaction methods have been proposed to collect self-reports of behavioral states through speech input, prompt users to verbally describe their ongoing posture or activity [37]. Research on large language models assists with context-aware personal health data, and health-related data reasoning [18, 38]. Recently, the large language models have been leveraged to support context-aware reasoning with personal health data, offering new possibilities for interpreting multimodal inputs and generating adaptive health insights [18, 38]. Also, the development of vision language models enables agents to understand the contextual information in images [39].

As XR technologies continue to evolve and become more widely accessible, current research has begun to investigate how XR systems can sense user context and environment to adapt system behavior. MR systems enable users to interact with the physical world through digital overlays while sensing contextual information to dynamically adapt to changing spatial and situational constraints [22]. A growing body of work focuses on adaptive spatial interfaces that adjust to real-world environments. InteractionAdapt preserves personalized interface layouts when users move between physical settings [12], while other work adapts MR interfaces based on semantic relationships between virtual content, physical space, and user activity [11, 39]. Adaptive systems have also been designed to respond to users' movement and workspace layouts [36], or to learn individual preferences from daily routines and optimize spatial layouts according to posture, spatial constraints, and visual field [40].

In summary, breaks and stretching have been shown to improve workplace health, and XR systems can augment exercise effectiveness through immersion and motivation. Recent advances in adaptive spatial computing further demonstrate how XR can sense context and adjust system behavior. However, workplace stretching systems rarely personalize interventions to users' daily ergonomic states. XR exercise systems mainly adapt based on in-system performance but overlook accumulated ergonomic context and the use of real-world supports. Adaptive spatial computing focuses on interface efficiency for daily tasks but has not been applied to ergonomics or physical activity. These gaps highlight the need for a system that integrates postural sensing and environment-aware adaptation to deliver stretches that are both contextually feasible and physically supportive.

3 FORMATIVE STUDY

To explore the challenges and opportunities in supporting effective workplace stretching for sedentary workers, we conducted a formative study involving interviews with clinical practitioners and an observation study and video analysis of common stretching practices, as Figure 2 shown. First, we interviewed clinical practitioners to understand current practices related to muscular fatigue, diagnosis, and treatment, aiming to validate and inform potential stretching design strategies. Next, we conducted an observational study to identify and prepare prompts

for VLM of possible ergonomic costs caused by workplace settings. Finally, we performed a video analysis of the stretching video to determine the final set of stretching exercises used in this study.

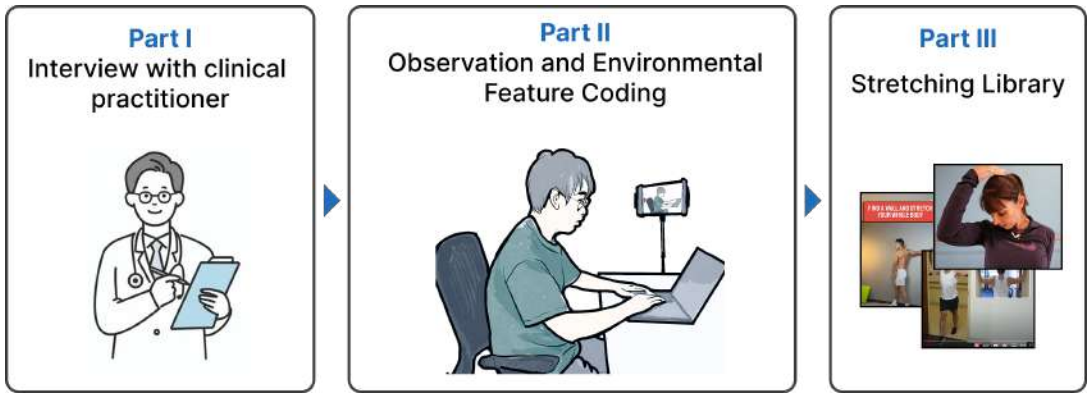


Fig. 2. The flow of formative study. We first conducted interviews with clinical practitioners to understand current practices related to muscular fatigue, diagnosis, and treatment. Next, we conducted an observational study to identify and prepare prompts for VLM of possible ergonomic costs caused by workplace settings. Finally, we performed a video analysis of the stretching video to determine the final set of stretching exercises used in this study.

3.1 Part I: Interview with Clinical Practitioner

Although prior work has documented common musculoskeletal issues among office workers and recommended rehabilitation strategies such as static stretching, it remains unclear how these principles are applied, prioritized, and evaluated in everyday clinical practice, particularly when exercises are prescribed for unsupervised home or workplace settings. We therefore interviewed clinical practitioners to ground insights from the literature in practical experience, clarify commonly treated problem areas, understand routine treatment and evaluation practices, and identify considerations relevant to remote rehabilitation. The findings informed the scope and design requirements of our system.

3.1.1 Participants. With the IRB approval, we conducted semi-structured interviews with five experienced residents and clinicians in a local hospital. Each had over three years of experience treating musculoskeletal disorders or rehabilitation.

3.1.2 Procedure. After we briefly introduced the study and gained consent from participants, we conducted a semi-structured interview. Interviews focused on common cases, treatment approaches, outcome evaluations, and challenges in home- and work-based rehabilitation. Each session lasted 30–45 minutes and was transcribed and thematically analyzed. Participants were compensated with local standards.

Common Cases and Causes. Participants reported that office workers frequently present with chronic pain in the neck, shoulders, and lower back, often following prolonged desk work, *"For example, patients reported that after long hours at the desk, the neck starts to hurt."* Neck and shoulder problems were described as the most prevalent and prone to recurrence *"For people with neck–shoulder issues, it tends to flare up easily,"* while prolonged sitting without postural change was said to more often cause lumbar discomfort *"people who sit for long periods usually have problems with the lower back."* The underlying mechanisms commonly involve tension in the upper trapezius and pectoralis minor *"In neck–shoulder cases, it's usually the upper trapezius and pectoralis minor that*

are tight" exacerbated by sustained forward-head and anterior-shoulder posture during computer use *"When using a computer, people tend to lean their head forward and keep their shoulders shifted forward."*

Treatment Practices. Standard rehabilitation focuses on guided static stretching of the anterior chain *"We stretch four areas: neck–shoulder and chest"*, often complemented by posterior chain stretching to improve support once the anterior muscles are lengthened. *"If the front side is opened up, it better supports the back."* Sessions typically begin with one-on-one instruction, during which clinicians demonstrate and correct technique until the patient achieves a clear stretch in the target muscle group. Clinicians noted that they sometimes perform manual stretching or physiotherapy for each other, *"Medical staff stretch each other's muscles or do therapy together"*, occasionally using specialized tools for targeted muscle release, *"We use specific tools to perform targeted stretching."*

Evaluation Strategies. In routine clinical practice, treatment outcomes are assessed primarily through subjective reports, *"Clinically, we rely on subjective assessment."* Clinicians ask whether the patient felt the target muscle group being stretched *"Did the target muscle group feel stretched?"* and whether there was noticeable improvement after a set, *"Was it better after one set?"* They may also note if the range of motion has increased *"Was the range of motion improved?"* but quantitative measurements are rarely employed outside research contexts.

Considerations for Remote Rehabilitation. For home or workplace programs, clinicians emphasize two key elements: adequate stretch range and sustained duration. They generally prefer static stretching held for 30 seconds to 2 minutes *"We tend to prefer static stretching, usually held for 30 seconds to 2 minutes"*, noting that *"You need to hold the muscle in the stretch for the effect to be better."* Clinicians further highlighted that static stretching is more suitable in unsupervised settings because it can be performed with external support, which improves stability and reduces risk. *"Static stretching allows people to rely on support, such as a wall or chair, so it is more stable and safer"* and *"With a fixed posture and support, it's easier to control the stretch and avoid overdoing it."* Clear and simple guidance is considered essential so that *"Even people without professional knowledge can do the stretch."*

In summary, effective stretching requires adaptation to individual conditions rather than fixed execution, as appropriate body targeting based on users' recent physical state and surrounding context. Formative interviews showed that in clinical practice, such adaptation is implicitly provided through professional judgment and feedback. However, in remote workplace or home-based settings, stretching is typically performed without real-time professional support, making it difficult for users to assess muscle status or ensure stable execution. Under these constraints, dynamic movements and dynamic stretching introduce additional safety concerns due to higher demands on motor control and difficulty in accurately localizing target muscles. Static stretching, by contrast, is clinically favored for its simplicity, lower execution variability, and clearer mapping between posture and target muscle groups, making it more suitable for unsupervised contexts. Nevertheless, fixed static stretching routines remain insufficient when divorced from users' accumulated working posture and immediate physical environment. These challenges motivate adaptive static stretching, which introduces individualized adjustment by sensing recent posture and spatial affordances to recommend stretches that can be safely and appropriately performed in situ.

3.2 Part II: Observation and Environmental Feature Coding for Prompt Generation

To identify key environmental variables that influence ergonomic risk and stretching feasibility, we conducted observations of real-world office environments. We targeted individuals whose daily routines involved prolonged sitting for desk-based tasks such as computer work. Participants were identified through campus-wide recruitment channels and screened to ensure that they met the criteria of spending a substantial portion of their workday seated. Participants include both graduate students and admin workers, this population is representative of sedentary workers, as they exhibit core characteristics of sedentary work patterns, including sustained sitting

durations, limited postural variability, and continuous engagement with screen-based tasks, which are commonly associated with musculoskeletal strain and reduced physical activity in modern office environments. The goal was to extract workspace and support features that could later be parameterized in the adaptive model, and derive a set of prompts for the vision language model in the system.

Table 1. Body Parts and Evaluated Ergonomic Features

Body Part	**Evaluated Features**
Neck	Presence, type, and positioning of headrests, neck pillows, or other supports; proper alignment for reduced cervical strain.
Upper/Lower Back	Degree of contact with the chair backrest; lumbar support adequacy; trunk posture and alignment.
Pelvis	Sitting centrally without asymmetrical pressure (e.g., from objects like wallets); allowance for hip movement to distribute load.
Shoulders	Armrest height and width; desk height; natural arm positioning; prevention of shoulder elevation or tension; input devices within reach.
Elbows/Forearms	Support from a desk or armrests; avoidance of resting on hard surfaces; presence of cushioning if needed.
Wrists	Keyboard angle; wrist rest usage; correct use means contact only during pauses, not during active typing.

3.2.1 Procedure and Data Collection. We conducted observations of 12 sedentary workers in their workplace for 30 minutes. Each 30-minute session was video-recorded to capture chair configuration, desk alignment, available support surfaces, and overall workspace openness. Observers annotated environmental factors whenever changes in participants' posture were observed, using a predefined checklist derived from prior academic literature and consultations with clinical experts [1].

3.2.2 Feature Definition and Prompt Schema. From these observations, we extracted the key features used to the prompt design to elicit reasoning from the VLM about ergonomic support, as Table 1 shown. Based on the coded data, we developed an initial set of prompts for the VLM to evaluate its capacity for reasoning about ergonomic support. These prompts were iteratively refined to better align with observed patterns and ergonomic principles. Finally, a clinical expert conducted a quality check on both the coding and the revised prompts to ensure accuracy and alignment with real-world ergonomic standards.

3.2.3 VLM Validation. To evaluate the performance of VLM, we developed a structured annotation codebook grounded in hallucination dimensions identified in prior literature [25]. Specifically, we assessed whether the VLM (1) correctly identified the key spatial relationship between the workspace and the human body, (2) produced rationales that violated task-specific evaluation rules, and (3) generated joint-level scores that deviated substantially from human judgment (defined as a difference greater than two points in this study).

We evaluated the model using an online public image dataset depicting seated working postures in office environments [26]. For each image, the VLM generated joint-level scores, accompanying rationales, and processing time. Two researchers independently annotated each joint-level output using a binary hallucination scheme (0 = no hallucination, 1 = hallucination) and provided human reference scores for each joint. Inter-rater reliability was assessed using Cohen's Kappa, yielding a value of 0.87, which indicates a high level of agreement.

Across all samples, the average processing time of the VLM was 16.98 seconds (SD = 8.78). At the joint level, the model achieved an overall accuracy of 86%, defined as the proportion of outputs without hallucination. In

addition, the mean difference between VLM-generated scores and human ratings was 0.93 (SD = 1.41), suggesting that while the model's scoring was generally aligned with human judgment, systematic deviations remained for certain joints and poses. Analysis of failure cases revealed that inaccuracies were strongly associated with environmental conditions, particularly lighting and occlusion. False negatives primarily occurred under poor lighting conditions, where shadows were sometimes misidentified as physical support structures (e.g., interpreting a shadow as a chair backrest), leading to incorrect inferences about the relationship between the chair and the torso. False positives mainly arose under occlusion caused by unfavorable viewing angles or intervening objects, where parts of the body or contact regions were partially or fully obscured.

3.3 Part III: Stretching Library

We conducted an iterative video search and screening process to identify commonly used stretching movements. Videos were reviewed until the observed stretching routines became highly repetitive and no substantially new movements emerged. The final dataset consisted of 46 videos, each video typically contained approximately 6–10 distinct stretching movements. All videos were sourced exclusively from verified official accounts to ensure reliability. We used the keyword "office stretch" and "static stretch", and included only verified or official channels to ensure content reliability. The initial pool contained a mix of dynamic exercises and full-body movements. To ensure consistency with our experimental focus, we applied four inclusion criteria: (1) the stretch must involve a static stretching (held position > 10 s) rather than dynamic repetition; (2) the primary movement should target the upper body; (3) the movement must be easily captured by the headset camera; (4) movements that were overly complex, risky, or yoga-based were excluded to maintain safety and feasibility.

Each selected action was coded with metadata including the targeted body parts, the required support surface (e.g., wall, table, doorframe, or none), and the necessary spatial configuration, as shown in Appendix B. After removing duplicates and filtering for suitability in this study, a total of 16 unique actions were included in the stretching action library. The final library covered a range of stretching actions such as wall-supported chest and shoulder stretches, table-based wrist flexion stretches, and free-space movements like neck stretches and trunk rotations, ensuring diversity across body regions and environmental contexts while remaining practical for XR guidance.

4 SYSTEM

We propose an Adaptive Stretching Pipeline (ASP), a modular and context-aware system that integrates posture sensing, ergonomic assessment, environment understanding, and adaptive action selection into an XR-based intervention framework.

4.1 Design Consideration

Drawing from formative study and prior research, we identify four key design considerations for developing effective stretching interventions in XR-based work environments:

Personalized Stretching Based on Behavioral and Postural Data. To effectively support musculoskeletal health, the system should adapt to each user by analyzing posture habits, activity patterns, and ergonomic stress points. Prior work highlights the value of tailoring exercise interventions to users' evolving physical needs [4, 40]. Posture-aware guidance can better target problem areas, replacing one-size-fits-all routines with meaningful, personalized movement.

Supported Static Stretching Duration and Supportive Objects. Correct form and timing are critical to the efficacy of static stretching. The interface must guide users to hold poses long enough. Clinicians often employ both manual stretching (where a therapist provides external force), mechanical stretching (using specialized devices in clinical contexts), and static stretching (holding a stretch for enough time and with a support object). However,

clinical practice and literature suggest that for the general public, self-stretching is the most appropriate and scalable method. In such cases, the availability of stationary supportive objects (e.g., walls, desks, or doorframes) is essential for alignment, balance, and posture maintenance [31]. Prior work has also demonstrated that aligning physical and virtual affordances in mixed reality (MR) exercise systems could shape effective and engaging experiences [3].

Progress-monitoring interactions. Real-time progress tracking can reduce ambiguity and improve motivation by providing users with clear goals. Dynamically adapting the appearance of augmented elements during and between exercises, such as changing visual indicators, can help users visualize completion and improvement [3, 41]. This feedback loop can enhance engagement and performance over time.

Guided Tutorials. Many users may be unfamiliar with proper stretching techniques. A brief, clear virtual tutorial can demonstrate correct form and help users build confidence before transitioning into active sessions. This aligns with prior recommendations for mixed-reality systems to align physical and virtual affordances for natural and safe movement [3].

4.2 System Overview

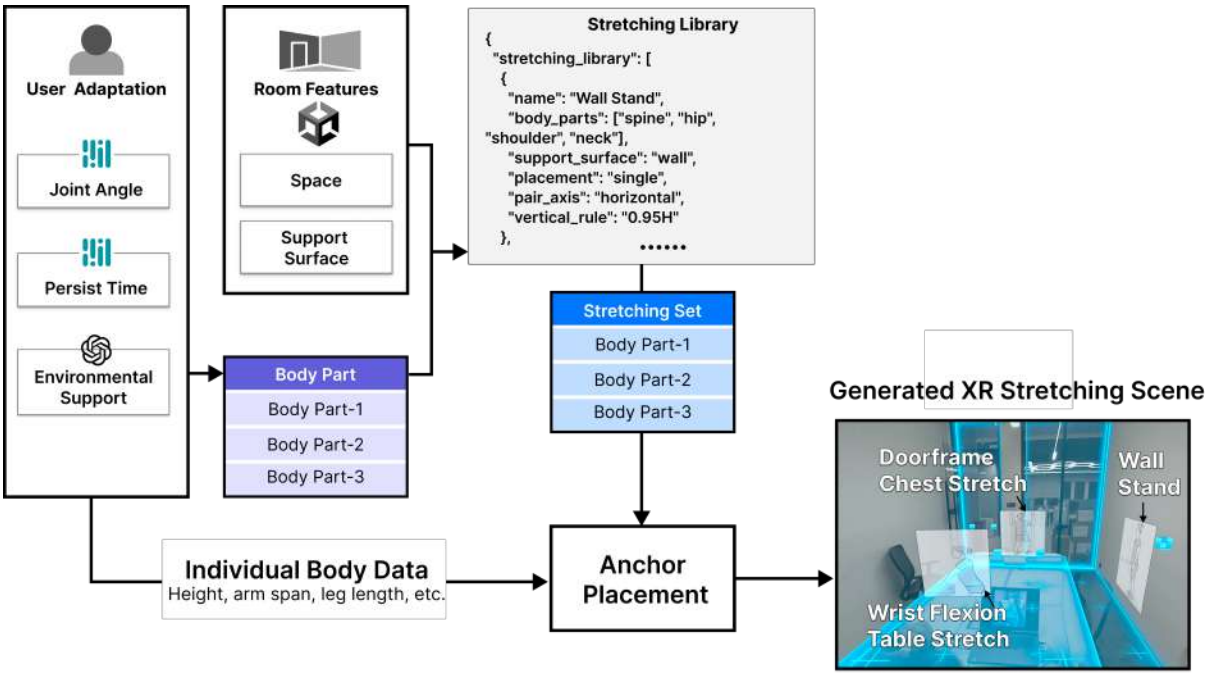


Fig. 3. Overview of the ASP Pipeline. The pipeline first captures joint angles and hold durations using MediaPipe, and combines this data with environmental analysis via a VLM to calculate a need score for each body region. Room features are extracted using the MRUK toolkit. Based on the need scores and room features, the system selects three stretching actions and adapts their positions to fit the user's workspace.

We propose an adaptive stretching pipeline, illustrated in Figure 3. The system first captures the user's posture during everyday work activities. From these data, it computes a *need score* for each body part by combining its deviation, persistence time, and the ergonomic cost caused by workplace setting. This process identifies which body regions are most in need of stretching.

Next, the system analyzes the workspace environmental features to detect feasible support surfaces (e.g., desk, wall, doorframe) using MRUK. By combining environmental features with posture-driven need scores, it selects suitable exercises from a predefined action library and assembles a personalized set of stretching actions.

Finally, the system instantiates interactive XR anchors aligned with the selected exercises. When a stretch begins, the anchors activate a countdown timer, and a progress indicator is displayed within the user's view.

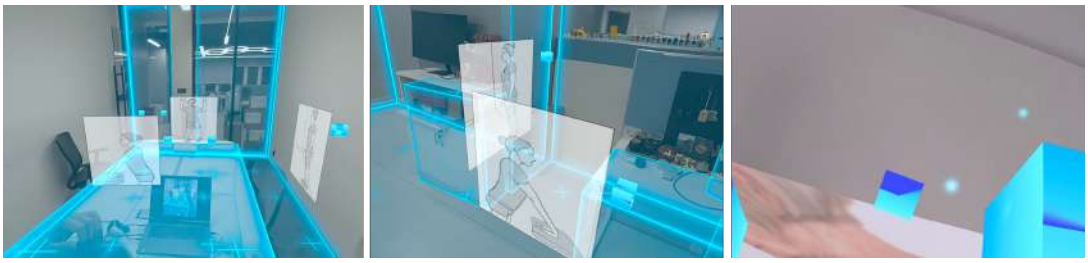


Fig. 4. Interaction Overview. (Left) Anchors generated in a meeting room. (Middle) Anchors generated in an open workspace. (Right) A progress indicator shows the stretching progress.

4.3 Stage 1 — Context Understanding

4.3.1 Ergonomic Data. To quantify ergonomic load, we compute a *need score* for each body part by combining posture risk and environmental support over a time window.

Table 2. Input parameters for adaptive stretching optimization.

Parameter	**Description**
User Posture Data	
$\mathcal{B} = \{b_{neck}, b_{wrist}, \dots\}$	Set of body parts (e.g., neck, wrist, back)
$\theta_b(t)$	Deviation angle of body part b at time t
$d_b(t)$	Dwell time of body part b (duration held in posture)
$m_b(t)$	Environment modifier for body part b (e.g., support from desk)
$S_b \in [0, 1]$	Need score of body part b in action a_i
Action Library	
$\mathcal{A} = \{a_1, \dots, a_n\}$	Library of candidate stretching actions
$\mathcal{B}(a_i) = \{b_{neck}, b_{wrist}, \dots\}$	Targeted body parts of action a_i (e.g., neck, wrist, back)
SupportSurfaceType$(a_i) \in$ {wall, doorframe, desk, none}	The required support surface for action a_i
RequiredSpace$(a_i) \in \mathbb{R}^3$	Required Space for a_i
Environment Context	
$\mathcal{R} = \{r_1, r_2, \dots, r_m\}$	Set of available anchors in the room
AnchorSurfaceType$(r_m) \in$ {wall, doorframe, desk}	Support surface type of anchor r_m
AvailableSpace$(r_m) \in \mathbb{R}^3$	Available free space around anchor r_m

Posture Score Computation. For each body part b, the instantaneous posture risk at time $t \in [0, T]$ is computed in four steps:

Angle Risk. For each body part b, two angular thresholds are defined based on ergonomic standards [1, 2, 42]:

- $\theta_{\text{low},b}$: the *middle risk threshold*. Deviations below this angle are considered low or negligible risk.
- $\theta_{\text{high},b}$: the *high risk threshold*. Deviations above this angle are considered high risk.

The instantaneous angle-based risk score $A_b(t)$ is then computed as

$$A_b(t) = \begin{cases} 0, & \theta_b(t) \le \theta_{\text{low},b}, \\ \dfrac{\theta_b(t) - \theta_{\text{low},b}}{\theta_{\text{high},b} - \theta_{\text{low},b}}, & \theta_{\text{low},b} < \theta_b(t) < \theta_{\text{high},b}, \\ 1, & \theta_b(t) \ge \theta_{\text{high},b}. \end{cases}$$

This formulation linearly normalizes the deviation angle into a bounded score $A_b(t) \in [0, 1]$. $A_b(t) = 0$ when the posture is within the safe zone ($\le \theta_{\text{low},b}$). $A_b(t)$ increases linearly between $\theta_{\text{low},b}$ and $\theta_{\text{high},b}$, representing gradual growth of risk. $A_b(t) = 1$ once the deviation exceeds $\theta_{\text{high},b}$, indicating high risk. This piecewise-linear definition provides a smooth transition between safe, moderate, and high-risk regions, aligning with ergonomic guidelines such as ISO 11226 and RULA, which classify posture into low, medium, and high risk levels.

Time Risk. To capture the effect of maintaining a posture over time, we define a dwell time $d_b(t)$, which accumulates while body part b continuously exceeds the middle-risk threshold $\theta_{\text{low},b}$.

The maximum tolerable duration decreases as the deviation angle increases:

$$T_{\text{max},b}(\theta_b) = \max\left(T_{\text{min},b},\ T_{0,b} \cdot \left[1 - \frac{\theta_b(t) - \theta_{\text{low},b}}{\theta_{\text{high},b} - \theta_{\text{low},b}}\right]\right),$$

where $T_{0,b}$ is the maximum tolerable time at $\theta_{\text{low},b}$, and $T_{\text{min},b}$ is the minimum tolerable time at $\theta_{\text{high},b}$. A linear decrease from $T_{0,b}$ to $T_{\text{min},b}$ provides a simple, interpretable approximation, indicates that larger deviations from neutral posture should be tolerated for shorter durations.

The instantaneous time risk is then

$$T_b(t) = \min\left(\frac{d_b(t)}{T_{\text{max},b}(\theta_b(t))},\ 1\right).$$

Notably, in this way, wealso ensure that $T_b(t) \in [0, 1]$. This ratio compares how long the body part has been held against how long it can safely be held at the current angle. It saturates at 1 once the posture exceeds its safe duration.

Table 3. body part thresholds and corresponding persistence times.

Body Part	θ_{low}	θ_{high}	T_0 (at low threshold)	T_{min} (at high threshold)
Neck	20°	45°	2 min	1 min
Shoulder	30°	60°	2 min	0.5 min
Forearm	60°	90°	3 min	1 min
Wrist	15°	30°	2 min	0.5 min
Trunk	20°	60°	2 min	1 min

Combined Instantaneous Risk. Angle and time risks are combined using a bounded union operator:

$$S_b(t) = A_b(t) + T_b(t) - A_b(t)\,T_b(t).$$

This calculation ensures $S_b(t) \in [0, 1]$, prevents double counting when both A_b and T_b are high, and reflects the ergonomic principle that either large deviation or long duration alone can create substantial risk.



Fig. 5. Sample output of the VLM environment-support analysis for posture. Red lines indicate body regions identified as high risk, yellow lines as medium risk, and green lines as low risk.

Ergonomic Cost of Workplace Setting. To incorporate physical support factors into the risk assessment of seated postures, we analyze the presence and adequacy of surrounding environmental elements, such as armrests, backrests, and neck supports using VLM, as figure 5 shown. We begin by setting up the model's context with a structured scoring framework, explicitly defining low-, medium-, and high-risk criteria across multiple posture components, including spinal alignment and limb support (see Appendix A for full scoring rubric). For each input image, typically a side-view photo of a seated individual, we prompt the VLM to assign discrete risk scores (0–10) to posture-specific elements like lumbar contact, elbow support, and wrist support.

With an environment adjustment factor $m_b(t) \in [-m_{\max}, m_{\max}]$ derived from semantic scene understanding (e.g., desk support, chair backrest), we set $m_{max} = 0.25$, the adjusted score is

$$\tilde{S}_b(t) = S_b(t) + m_b(t),$$

Aggregation Over the Time Window. The final posture score for body part b is obtained by time-averaging:

$$\bar{S}_b = \frac{1}{T} \int_0^T \tilde{S}_b(t)\, dt.$$

4.3.2 Spatial Information. We leverage the Meta Reality Utility Kit (MRUK) to analyze the spatial context of the user's workspace. As figure 4, MRUK detects a set of available anchors

$$\mathcal{R} = \{r_1, r_2, \ldots, r_m\},$$

where each anchor r is characterized by two attributes:

$$\text{AnchorSurfaceType}(r_m) \in \{\text{wall}, \text{desk}, \text{doorframe}\}, \quad \text{AvailableSpace}(r_m) \in \mathbb{R}^3.$$

The *anchor surface type* specifies the category of physical structure (e.g., wall, desk, doorframe), while the *available space* denotes the maximum unobstructed region adjacent to that anchor. For example:

- **Wall:** maximum continuous length of the wall unobstructed by objects
- **Desk:** free area around desk edges, including usable desk height
- **Doorframe:** free region surrounding the doorframe without blocking elements

From each anchor, we compute $\text{AvailableSpace}(r_m)$ and extract its 3D center point, which serves as a spatial feature for filtering feasible actions in the optimization stage.

4.4 Stage 2 — Action Set Optimization

4.4.1 Release Score Definition. Each candidate action $a_i \in \mathcal{A}$ targets a subset of body parts $\mathcal{B}_i \subseteq \mathcal{B}$. Its release score is defined as the sum of the current need scores of the targeted body parts:

$$R_i = \sum_{b \in \mathcal{B}_i} \bar{S}_b.$$

4.4.2 Feasibility Constraint. An action a_i is eligible only if there exists an anchor $r_m \in \mathcal{R}$ in the room such that both its support surface type and available space satisfy the requirements of a_i:

$$\text{SupportSurfaceType}(a_i) = \text{SupportSurfaceType}(r_m) \quad \wedge \quad \text{RequiredSpace}(a_i) \leq \text{AvailableSpace}(r_m).$$

4.4.3 Selection Procedure. We select a set of three actions $\mathcal{S}^*$ using the following iterative procedure:

(1) Compute R_i for all feasible actions.
(2) Select the action $a^* = \arg\max_i R_i$ and add it to $\mathcal{S}^*$.
(3) For each body part $j \in \mathcal{B}_{a^*}$, update its need score by halving it:
$$\bar{S}_j \leftarrow \tfrac{1}{2}\bar{S}_j.$$
(4) Recompute release scores for all remaining actions and repeat until $|\mathcal{S}^*| = 3$.

This procedure prioritizes actions that maximize overall release while progressively reducing the dominance of any single body part to promote diversity, and it achieves this in linear time with algorithmic complexity $O(n)$, offering a clear efficiency advantage over exhaustive search.

4.5 Stage 3 — Anchor Placement

Each selected action a is instantiated from a declarative spec in the stretching library:

$$\textsc{Spec}(a) = \big(\textsc{Anchor}(a),\ \textsc{Placement}(a),\ \textsc{PairAxis}(a),\ \textsc{PairOffset}(a)\big)$$

These attributes determine both the *surface-level* feasibility (which anchor type is valid) and the *geometry-level* placement (how many anchors, where, and how they are oriented).

4.5.1 Anchor Matching. For each a, MRUK provides candidate surfaces C_a with center c and local frame. Candidates must satisfy:

$$\textsc{Type}(c) = \textsc{Anchor}(a), \qquad \textsc{Space}(c) \geq \textsc{MinSpace}(a).$$

4.5.2 Anchor Placement. Let $c = (c_x, c_y, c_z)$ be the center of the chosen anchor surface. Anchors are placed by adding offsets along the axes determined by the stretching library:

$$\mathbf{p} = (c_x,\ c_y,\ c_z) + axisoffset$$

4.6 Implementation

We implemented the adaptive stretching pipeline in Unity 2022.3.34f1 with MRUK v74. Room feature data were captured directly on the headset using the Oculus Room features and processed with MRUK. Postural data were obtained through MediaPipe, which tracked joint angles and dwell times. To incorporate environmental support into the need score computation (e.g., armrests, backrests, neck supports), we analyzed the surrounding objects using GPT-4o vision–language model (VLM).

5 EVALUATION

This study investigates how the adaptive XR stretching program (ASP) can support effective and acceptable stretching breaks during work while accounting for varying levels of perceived social exposure. Through a

controlled experiment comparing ASP with a commercial stretching application (Bend) and a randomized XR baseline across different social environments, we examine how adaptive support influences perceived time, comfort, effort, and social discomfort during workplace stretching.

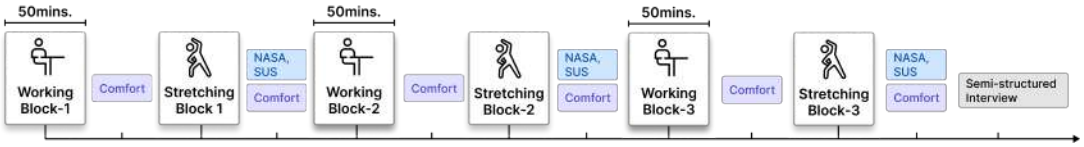


Fig. 6. Study procedure. After 50 minutes of work, perceived comfort were assessed prior to stretching. They then performed the first stretching session, followed by measurements of perceived comfort, and completion of the NASA-TLX and SUS questionnaires. The same procedure was repeated for Working Block 2 and 3.

5.1 Study Design

The study comprised three experimental blocks and evaluated the performance of ASP by comparing it with two baselines: an off-the-shelf stretching application (Bend) and a randomized XR stretching condition. Bend is a commercial mobile stretching application designed to support general flexibility and mobility training. It provides a wide range of stretching exercises targeting different body regions, including static stretches and dynamic movements. In this study, we primarily used its static stretching content to match the experimental focus on low-intensity workplace stretching. Each stretching exercise in Bend is presented with a combination of textual instructions and animated demonstrations, as figure 11 shown.

As figure 6 shown, each block consisted of 50 minutes of desk-based work followed by a 5-minute stretching session. In the Bend condition, the researcher first filtered relevant stretching movements from the application's library, after which participants selected stretch sets based on their personal preference. In the randomized XR condition, stretch sets were automatically selected at random from the same stretching library.

We employed a within-subjects design in which each participant experienced all three conditions under varying social exposure. The experiment further manipulated perceived social exposure across three environments. Perceived social exposure was systematically varied by introducing visual and attentional social cues, drawing on social presence and social translucence frameworks. Environment 1 contained no bystanders, representing a private meeting room or private office. Environment 2 included bystanders engaged in unrelated tasks and showing no attention toward the participant. Environment 3 included bystanders who occasionally directed their attention toward the participant. Environments 2 and 3 were conducted in a controlled open workspace; bystanders in this workspace were instructed not to engage with or pay any attention to the participants. In Environment 3, additional researchers, who were unknown to the participants, acted as bystanders and occasionally directed attention toward participants during stretching. The order of stretching conditions and environments was counterbalanced. We measured perceived time, perceived comfort, and subjective outcomes including NASA-TLX, SUS, and social discomfort.

5.2 Participants

We recruited 15 participants from a local university include both graduate students and admin workers (7 male, 8 female; age: Avg = 26.47, sd = 2.26, 9 participants were participated in the previous formative study). All participants reported having used AR/VR devices before, with varying familiarity. All of the participants have the need of long-term sedentary work.

5.3 Procedure

After a brief introduction and obtaining informed consent, a phone camera was then set up to capture posture, and participants continued their usual work activities. During this time, researchers used the Oculus Room feature to scan the workspace.

After 50 minutes of work, participants completed a pre-stretching comfort questionnaire. After completing each action, participants reported their perceived duration. This approach was intended to reduce participants' awareness of being observed and encourage more natural interaction with the system. Following the stretching session, comfort levels were recorded, along with the NASA-TLX, SUS, and social acceptability questionnaires for the given condition. The same procedure was repeated for the other condition. Finally, after completing both blocks, a semi-structured interview was conducted with each participant. Participants were compensated with local standards.

5.4 Result

5.4.1 Joint-Level and Overall Comfort Changes. To assess overall comfort change across joints, joint-level improvements were summed for each participant and method, as figure 7 shown. A one-way ANOVA revealed a significant main effect of method ($p < .001$). Post-hoc paired comparison shown overall comfort improvement was significantly greater for the adaptive method compared to both the bend method ($p < .001$) and the random method ($p < .001$). The bend method also outperformed the random method ($p < .001$).

Joint-level comfort changes were analyzed using an ordinary least squares regression model with participant-clustered standard errors to account for repeated observations within participants. The model included the main effects of interaction method and joint, as well as their interaction. Significant method-by-joint interaction effects were observed at the wrist. Both the bend method with mean=0.6 ($p = .001$) and the random method with mean=0.23 ($p = .031$) resulted in significantly smaller comfort improvements at the wrist compared to the adaptive method with mean=1.6. No other interaction terms reached statistical significance, although several joints exhibited consistent directional trends.

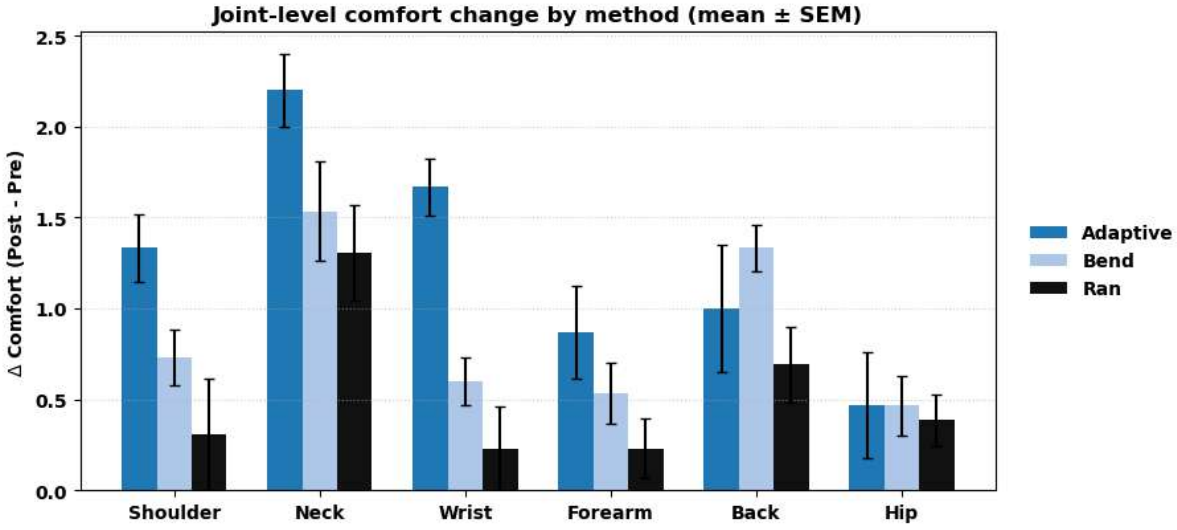


Fig. 7. Δ Preceived comfort level for each body part after stretching.

5.4.2 Social Acceptancy. We analyzed the data using linear mixed-effects models with participant as a random intercept. Social context (three levels) and interaction method (three types) were treated as within-subject fixed effects. The mixed-effects model revealed a strong main effect of social context on perceived social acceptability. Descriptively, social discomfort increased monotonically with social exposure (env 1: M = 1.53, SD = 0.64; env 2: M = 3.80, SD = 1.01; env 3: M = 5.13, SD = 2.39). Compared to the lowest social exposure level, both env 2 ($p < .001$) and env 3 ($p < .001$) social exposure levels resulted in significantly higher social discomfort ratings.

There was also a main effect of the interaction method. The bend condition showed slightly higher social discomfort (M = 4.27, SD = 2.49) compared to the ASP (M = 3.20, SD = 1.74; p = .046), whereas the random condition (M = 3.00, SD = 2.00) did not significantly differ from ASP.

P5 shared *"With the headset on, people might think I'm doing something professional or gaming, not just stretching"* Similarly, P1 noted that stretching without a headset heightened self-consciousness *"When someone is looking at me, I really want to explain I'm just stretching."* The immersive elements also helped distract from social discomfort: *"I was focused on the countdown and didn't even notice people around me."*

Participants suggested masking or replacing bystanders in the environment: *'"I'd feel more comfortable if I didn't see people watching me, maybe replace them with non-human characters."*

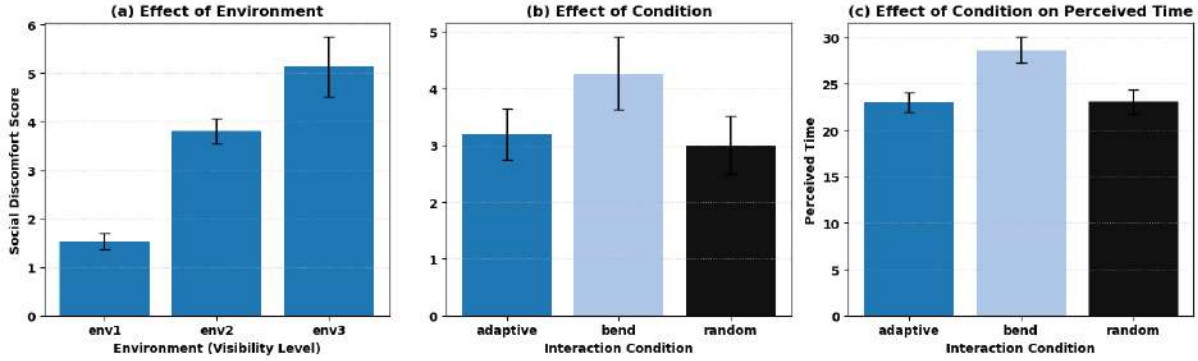


Fig. 8. Result of Social Discomfort and Perceived Time. a) Effect of environment on social discomfort. b) Effect of condition on social discomfort. c) Perceived Time across Interaction condition

5.4.3 Perceived Time, Engagement, and Motivation. Perceived time was analyzed using a repeated-measures ANOVA with the interaction method as a within-subject factor. The analysis revealed a significant main effect of interaction method on perceived time ($p < .001$). The bend condition was perceived as taking longer ($M = 28.67, SD = 5.50$) than both the adaptive ($M = 23.00, SD = 4.14$) and random ($M = 23.07, SD = 5.12$) conditions. Post-hoc paired comparisons showed that perceived time in the bend condition was significantly longer than in the ASP ($p < .001$) and random ($p < .001$) conditions, while no significant difference was observed between the adaptive and random conditions.

This reduction in perceived time was echoed in participants' comments. Many described the experience as more immersive and engaging, reframing stretching as an interactive task rather than a passive activity *"It was interesting, stretching became the interaction trigger,"* and *"It felt like completing a mission in a game."* Several participants noted that ASP helped them stay focused, making the session feel less effortful and less time-consuming *"In daily stretching, I just do 10 seconds and give up. But here, I focused on finishing the mission and forgot about time. It felt more effective."* Beyond immersion, ASP also increased motivation to initiate and sustain stretching breaks, as one participant explained,*"It reminds me of the one-minute stand alert on my wristband, but*

I usually ignore that. Here, there's something to interact with, so I actually want to do it." In the Bend condition, participants typically placed the phone nearby, but it was not always within their field of view. As a result, they had to repeatedly shift their attention to the phone to check their progress, which disrupted the experience. As one participant commented, *"I can't see the progress, so I need to check it frequently. It feels like it's taking much longer."*

Participants also suggested ideas for increasing engagement further, such as stronger visual effects and light gamification *"More salient visuals would improve achievement"*, *"You could avoid obstacles when doing neck stretches"*, *"Maybe push elements forward during chest stretches."*

5.4.4 Subjective Workload and Usability. Subjective workload and usability were assessed using NASA-TLX and SUS (Fig. 9; detailed descriptive statistics are reported in Appendix). Overall, the adaptive condition consistently showed lower perceived workload and higher usability than both the random and bend conditions.

Across NASA-TLX dimensions, participants reported lower mental, physical, and temporal demand, as well as reduced effort and frustration in the adaptive condition. Participants attributed this reduction primarily to the system's ability to recommend suitable stretches and guide positioning, which reduced the need for decision-making and environmental assessment. As one participant noted, *"It helps me save one step. I don't need to search or think about whether the environment has enough space to stretch."* In contrast, participants in the bend condition reported higher temporal demand and effort due to the need to repeatedly shift their gaze to an external device to check progress, whereas MR-based guidance kept progress information within the field of view.

SUS results further indicated higher willingness to use, perceived intuitiveness, and integration into the work context for the adaptive condition, alongside lower perceived complexity and higher confidence compared to the baselines. While confidence was generally higher, some participants noted reduced confidence when system recommendations did not align with personal needs, for example, *"I usually stretch my back after sitting for a while, but the recommended stretches didn't target it."* Ratings of required prior knowledge were similar across conditions, suggesting comparable accessibility for users with limited stretching experience.

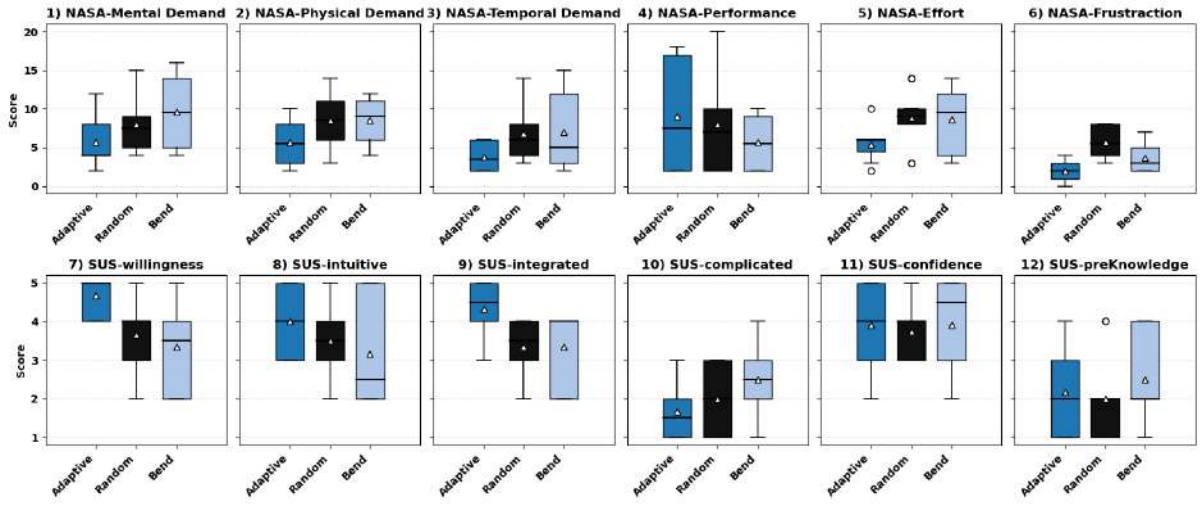


Fig. 9. NASA-TLX and SUS result.

6 DISCUSSION AND DESIGN IMPLICATION

In this work, we designed an Adaptive Stretching Pipeline (ASP) that personalizes brief static stretches from posture history and environmental affordances. We conducted a formative study to specify the pipeline and a controlled user study to evaluate it against a commercial application (Bend) and randomised condition under a controlled social exposure environment. Below, we synthesize key findings and then translate them into concrete design implications.

6.1 Summary and Discussion

6.1.1 Personalized Stretching Improves Comfort. Our results show that ASP improved perceived comfort in several body parts, however, statistically significant differences were observed only at the wrist, while other body regions showed non-significant positive trends. This pattern indicates that posture- and environment-aware adaptations can effectively target body regions that are commonly overlooked in self-directed stretching and help compensate for asymmetrical or cumulative load during prolonged sedentary work. ASP consistently outperformed both user-preferred (bend) and random stretching in comfort- and relief-related outcomes. While allowing users to select stretches based on preference led to moderate improvements over random assignment, its benefits remained limited, likely due to users' incomplete or inaccurate awareness of their own bodily strain, especially in regions with low perceptual salience.

By contrast, ASP leverages context information to form a closed loop of objective assessment. This enables the system to proactively address latent tension in regions that users may not consciously identify as problematic. This finding aligns with prior work showing that personalization enhances adherence to physical activity interventions [4], and further extends it by demonstrating that the accuracy of personalization plays a decisive role in therapeutic outcomes.

6.1.2 Decision Making. Consistent with the lower mental demand and effort reported in the NASA-TLX results, ASP reduced users' mental load by automatically recommending appropriate actions and guiding execution. However, these benefits also revealed a design trade-off. Despite lower workload and higher usability overall, some participants, particularly those with existing discomfort or established stretching routines, felt that automated recommendations limited their sense of autonomy when they did not align with personal needs. This suggests that reducing cognitive effort through automation does not necessarily translate to a stronger sense of control. Several participants expressed a desire to understand the rationale behind specific stretch recommendations and to have the ability to adjust them when necessary. Prior work similarly suggests that explainability can foster user trust and confidence in performing system-guided tasks [34, 62].

6.1.3 Motivation through Interaction and Immersion. Perception substantiated by the quantitative finding that XR displays (ASP and Randomised condition) significantly reduced perceived time compared to Bend. Prior work in cognitive psychology suggests that distorted time perception is an indicator of attentional engagement and intrinsic motivation, activities that feel shorter than they are tend to be perceived as less effortful and more repeatable [60].

This interpretation is further supported by participants' qualitative accounts, in which ASP was described as more immersive and goal-oriented than static tutorials. The interactive mechanics and task-based structure appear to redirect attention away from time monitoring toward action completion, aligning with goal-setting theory and prior findings on gamification and XR engagement [41]. Several participants spontaneously contrasted ASP with smartwatch-based break reminders, which they characterized as monotonous and easy to ignore over time. This comparison resonates with existing literature on just-in-time interventions, which often suffer from habituation and declining effectiveness due to low perceived reward and minimal engagement [48].

6.1.4 Stretching in Shared Spaces. The results suggest that perceived social discomfort during stretching was strongly influenced by social exposure and condition. As social exposure increased from Env1 to Env3, participants reported higher levels of social discomfort, indicating that perceived observation may contribute to shaping users' responses during physical activity. Across conditions, the Bend condition elicited the highest level of social discomfort, indicating that overt, posture-focused stretching movements can feel socially awkward regardless of context. In contrast, ASP and the randomized MR condition resulted in comparable levels of reported discomfort. This pattern may be explained by changes in how participants perceive their surroundings. In MR-based conditions, immersive engagement may shift attention away from monitoring the social environment, thereby reducing users' awareness of being observed, this suggests that MR alters how users perceive and attend to the surrounding social context.

6.2 Design Considerations

Based on these findings, we propose six key guidelines for designing adaptive stretching systems:

6.2.1 Integrate Diverse Input Sources. Beyond posture and ergonomic data, participants emphasized the importance of incorporating personal factors such as clothing choices and historical discomfort records. For instance, certain outfits restricted large movements, which shaped participants' preferences for smaller or less conspicuous stretches. Similarly, users with long-term pain in specific body regions (e.g., lower back) expected the system to prioritize those areas. Adaptive systems should therefore integrate both contextual and personal data to refine stretch recommendations.

6.2.2 Make Adaptation Explainable. Adaptive stretch recommendations reduced participants' mental demand by relieving them of the need to decide which stretches to perform, thereby lowering cognitive effort and decision-making load. However, when participants lacked clarity about why certain stretches were suggested, such as which body regions were targeted or how recommendations related to their posture history, they experienced residual cognitive tension or uncertainty. This gap in transparency can undermine trust and engagement, as users may hesitate to follow recommendations they do not understand. Providing explanations or visually highlighting targeted body regions can therefore not only maintain the reduction in mental demand but also enhance informed participation and perceived control over the exercise routine [34, 62].

6.2.3 Adjust Guidance to Skill Level. Tailor the level of guidance to the user's expertise. Novice users benefit from step-by-step support, while experienced users may prefer minimal prompts. As prior research suggests, adaptive guidance improves usability and engagement across skill levels [3]. Also, provide clear, multi-modal guidance that combines visual anchors with complementary cues such as voice instructions, avatar demonstrations, or virtual hand gestures. These supports help improve user confidence and learning efficiency, especially for novices who may lack prior experience with stretching or physical exercises.

6.2.4 Use Playful and Interactive Elements. The significant reduction in perceived duration observed in ASP suggests that interactive XR-based stretching can lower users' subjective effort even when the physical demands and duration remain constant. For short, low-intensity exercises such as workplace stretching, perceived effort is a critical determinant of whether users are willing to initiate and repeatedly engage in the activity. Designing interactions that make stretches feel shorter and less burdensome can therefore play a central role in supporting long-term motivation.

Gamified and immersive elements, such as task-oriented mechanics, obstacle avoidance, or simple action-based interactions (e.g., punching or pushing), can reframe stretching from a passive, effortful routine into an active goal-driven task, thereby reducing perceived effort. Participants' preference for salient visual feedback further suggests that reinforcing a sense of progress or achievement may strengthen intrinsic motivation to repeat the

activity [30, 54]. Importantly, these elements should be implemented within a flexible design space that allows designers to tune the level of playfulness to different users and contexts.

At the same time, designers must carefully balance immersion with movement quality and ergonomic safety. Excessive visual stimulation or overly complex game mechanics may distract users from maintaining correct posture or movement execution, potentially reducing the effectiveness or safety of stretching routines. Effective designs should therefore leverage gamification primarily as a means to reduce perceived effort and motivational barriers, while preserving clear guidance and accuracy in physical movement.

6.2.5 Design for Social Awareness. Our results indicate that social discomfort was lower in Env2 (public but unattended) than in Env3 (public with others' attention), highlighting that perceived observation, rather than publicness alone, drives social discomfort. Although MR-based adaptive and random conditions showed comparable discomfort levels and both were lower than bend-based stretching, increased social exposure consistently elevated discomfort. Participant feedback aligns with this pattern, while the headset reduced self-consciousness, discomfort persisted when participants felt watched. Therefore, adaptive stretching systems should explicitly address perceived social attention, for example, by masking or abstracting bystanders, filtering them from the field of view, or reframing them as non-human elements to reduce the sense of being observed in highly exposed environments.

6.2.6 Design for multi-user stretching. Future adaptive exercise systems could extend beyond individual and environmental awareness to incorporate social context. During the formative study, clinical practitioners noted that stretching is often performed collaboratively, where partners assist each other in maintaining posture or achieving a proper range of motion. Building on this observation, future MR exercise systems could support co-located collaborative stretching among co-workers, enabling mutual motivation and social accountability. Such systems may promote positive workplace culture by transforming individual recovery breaks into shared, socially engaging activities. Designing for social acceptability, synchronization, and group dynamics will be essential to ensure comfort and inclusivity, particularly in open or shared office environments. Integrating social interaction into adaptive stretching could therefore enhance long-term engagement and lower psychological barriers to participation.

6.3 Limitation and Future Work

This work has several limitations that point to opportunities for future research.

First, the stretching library was limited in both variety and scope. We primarily focused on upper-body stretches to address common sedentary discomfort in office settings and to accommodate the tracking limitations of current camera and headset technologies. Future systems should expand to include lower-body and full-body stretches, particularly by adopting more diverse sensing modalities beyond vision-based tracking.

Second, our evaluation was limited to a short-term, controlled study. While initial results are promising, longer-term deployments are necessary to understand how adaptive stretching systems affect behavior, comfort, and engagement over time. A longitudinal study could reveal patterns of adherence, changes in user preferences, and potential fatigue or novelty effects. Moreover, to mitigate potential Hawthorne effects, researchers were not actively observing participants during the stretching phase, but acted as passive bystanders, and participants completed the stretching tasks independently without direct supervision. This aimed to reduce participants' awareness of being observed and encourage more natural engagement with the system. Nevertheless, the Hawthorne effect cannot be entirely eliminated in short-term laboratory studies, as participants may still modify their behavior due to the novelty of the system or the experimental setting itself. A longitudinal deployment could further attenuate this effect by allowing participants to habituate to both the system and the study context

over time, thereby better capturing sustained usage patterns, motivation, and adherence under more ecologically valid conditions.

Moreover, the current evaluation adopted a fixed schedule for breaks. Prior research has emphasized the importance of appropriate timing in break and intervention delivery, noting that inopportune or poorly timed prompts can lead to distrust or disruption [13, 48]. Future development of adaptive micro-break systems should therefore consider when and how to notify users, tailoring the timing of interventions to individual work rhythms and contextual states. Advances in sensing and context-aware technologies may support this goal, for instance, by detecting users' work progress, posture, or physiological indicators of cognitive engagement to deliver timely and personalized micro-break experiences.

While MR provides an engaging and immersive experience, the requirement to put on a headset specifically for a short microbreak introduces a nontrivial interaction cost. In real-world workplace contexts, this physical and temporal friction may outweigh the benefits of the exercise itself, potentially discouraging users from participating. This tension highlights an important design trade-off, while MR can enhance motivation and embodiment, the logistical overhead associated with device use may limit adoption, particularly for brief, frequent interventions like microbreaks. Future work should consider strategies to minimize this burden, such as lighter wearable devices, integration with existing hardware, or context-aware just-in-time activation to align engagement benefits with practical usability.

Future work could explore strengthening the technical novelty of the system by moving from rule-based adaptation toward more principled and personalized computational models. The current linear need score could be extended into a hybrid or learned user-adaptive model that incorporates individual differences in physical capacity, fatigue accumulation, and compliance behavior, enabling personalization beyond population-level ergonomic thresholds. In addition, the greedy scheduling strategy may be generalized into a multi-objective optimization framework that explicitly balances biomechanical load, temporal disruption, spatial feasibility, and long-term adherence under real-time constraints. Beyond reactive adaptation, future systems could model posture trajectories and work rhythms to support predictive and anticipatory interventions, shifting from responding to accumulated strain toward preventing it.

Future work should also broaden the application context. The current system is designed for low-mobility, desk-bound environments, but many occupations involve different ergonomic risks. Stretching interventions could be adapted for physically demanding roles, such as cleaners, security personnel, or retail workers who face prolonged standing, repetitive motions, or constrained postures.

Finally, the posture recognition phase raised privacy concerns, particularly in shared or professional settings, though vision-based posture recognition has demonstrated high accuracy [33]. Additionally, camera use may be prohibited in sensitive workplaces. These concerns are further amplified when workspace imagery is processed via cloud-based services, which may inadvertently capture sensitive on-screen content or non-consenting bystanders. While the current prototype adopts cloud-based analysis for exploratory purposes, this highlights the importance of both alternative sensing modalities (e.g., wearable sensors) and privacy-preserving approaches (e.g., on-device processing). In addition, interaction-level mitigation strategies could be deployed that allow users to retain control over data capture, such as user-triggered sensing and shortcut interactions to temporarily pause monitoring during privacy-sensitive moments.

7 CONCLUSION

This work introduced the Adaptive Stretching Pipeline (ASP), a mixed reality system that personalizes brief workplace stretches based on users' posture history and environmental affordances. By combining posture tracking with scene understanding, ASP adapts both the content and context of movement interventions. Our user study provides initial evidence that such posture-aware and context-aware adaptation can influence users' experience

of workplace stretching, particularly in terms of perceived effort, engagement, and time perception, compared to non-adaptive baselines. Based on our findings, we proposed several design considerations for future adaptive movement systems, including the need for explainable adaptation, multimodal guidance, gamified interaction, social context awareness, and privacy-conscious sensing. While ASP focused on upper-body movements in sedentary settings, future work should expand the stretching library, support long-term use, address diverse workplace contexts.

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Appendix A: VLM Prompt for Postural Risk Assessment

Evaluation Criteria

Overall Scoring Logic. Each assessment item is rated on a scale from Low Risk, Medium Risk, to High Risk. The scoring ranges are:

- Low Risk: 0–2 points
- Medium Risk: 3–6 points
- High Risk: 7–10 points

The total risk score is the sum of all item scores.

Part 1: Chair Backrest and Lumbar Support Evaluation Criteria: Back Support and Posture Focus: Assess how closely the user's back contacts the chair (contact area and gaps), especially around the lumbar spine.

Scoring Guide: Low Risk (0–2): Full, close contact between back (including lumbar area) and the chair. Lumbar support aligns well, trunk is upright or slightly reclined (100–110°), posture relaxed. Medium Risk (3–6): Partial contact with gaps. Lumbar support is misaligned or posture leans slightly forward. High Risk (7–10): Significant forward lean or slouching. Lumbar area unsupported or poorly used.

Part 2: Arm and Shoulder Support

Subunits: Left Shoulder, Right Shoulder, Left Arm, Right Arm

Shoulder Posture (Left/Right) Focus: support alignment.

Scoring Guide: Low Risk (0–2): Neutral, relaxed posture. Arm well-supported. No overreach. Medium Risk (3–6): Occasional tension or misalignment. Support slightly off. High Risk (7–10): Prolonged shrugging or slouching. No effective support.

Arm Posture (Left/Right) Focus: forearm support, and surface pressure. Scoring Guide: Low Risk (0–2): Full support. Medium Risk (3–6): partial support. High Risk (7–10): no support, or prolonged surface pressure.

Wrist Posture (Left/Right) Focus: use of wrist rests. Scoring Guide: Low Risk (0–2): Wrist aligned, rest used only during pauses. Medium Risk (3–6): Minor deviations or light contact during typing. High Risk (7–10): Sustained misalignment or pressing on edges.

Neck Posture Focus: Check for neck support Scoring Guide: Low Risk (0–2):Proper neck support Medium Risk (3–6): ineffective support use. High Risk (7–10): Poor alignment

Instructions Carefully analyze the user's seated posture from the provided side-view image. Score each of the body regions using the above criteria. Provide a brief visual justification for each score. Calculate the total score and determine the overall risk level.

Output Format Return the results in the following JSON structure:

```
{
  "assessment_summary": {
    "total_risk_score": <sum of all sub-scores>,
    "overall_risk_level": "<Low Risk / Medium Risk / High Risk>"
  },
  "detailed_assessment": [
    {
      "item_id": "trunk",
      "item_name": "Back Support and Posture",
      "risk_level": "<Low Risk / Medium Risk / High Risk>",
      "score": <0-10>,
      "reasoning": "<Justification>"
    },
```

```
    ... (repeat for other body regions) ...
  ]
}
```

Appendix B: Stretching Library

```
{
  "stretching_library": [
    {
      "name": "Wall Stand",
      “body_parts": ["spine", "hip", "shoulder", "neck"],
      "support_surface": "wall",
      "placement": "single",
      "pair_axis": "horizontal",
      "vertical_rule": "0.95H"
    },
    {
      "name": "Wall Side Stretch",
      "joints": ["hip", "shoulder"],
      "support_surface": "wall",
      "placement": "symmetric_pair",
      "pair_axis": "vertical",
      "vertical_rule": "0.85H"
    },
    {
      "name": "Wall Chest Stretch",
      "joints": ["shoulder"],
      "support_surface": ["wall", "doorframe"],
      "placement": "single",
      "pair_axis": "horizontal",
      "vertical_rule": "0.80H"
    },
    {
      "name": "Standing Chest Stretch – Wall",
      "joints": ["shoulder", "elbow"],
      "support_surface": "wall",
      "placement": "single",
      "pair_axis": "horizontal",
      "pair_offset": 0.35,
      "vertical_rule": "0.85H"
    },
    {
      "name": "Doorframe Chest Stretch (Forearm)",
      "joints": ["shoulder"],
      "support_surface": "doorframe",
      "placement": "symmetric_pair",
```

```
      "pair_axis": "horizontal",
      "vertical_rule": "0.82H"
    },
    {
      "name": "Neck Stretches",
      "joints": ["neck", "shoulder"],
      "support_surface": "none",
      "placement": "single",
      "pair_axis": "horizontal",
      "vertical_rule": "0.95H"
    },

    ......
  ]
}
```


Appendix C: Descriptive Statistic of NASA-TLX and SUS Questionaire

Table 4. Descriptive statistics (Mean, SD) for SUS and subjective usability ratings across conditions.

Measure	Adaptive	Random	Bend
Willingness	4.67 (0.49)	3.67 (0.98)	3.33 (1.15)
Intuitive	4.00 (0.85)	3.50 (1.00)	3.17 (1.40)
Integrated	4.33 (0.78)	3.33 (0.78)	3.33 (0.98)
Complicated	1.67 (0.78)	2.00 (0.85)	2.50 (1.00)
Confidence	3.92 (1.00)	3.75 (0.75)	3.92 (1.24)
Pre-knowledge	2.17 (1.27)	2.00 (1.04)	2.50 (1.17)

Appendix D: Video Filtering Process

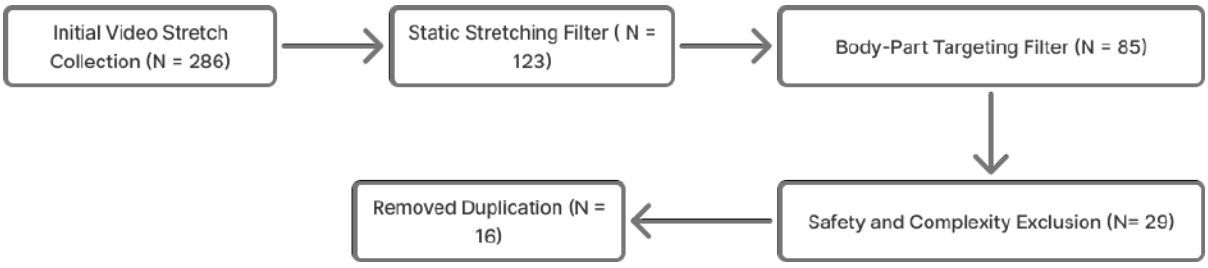


Fig. 10. Video Filtering Process

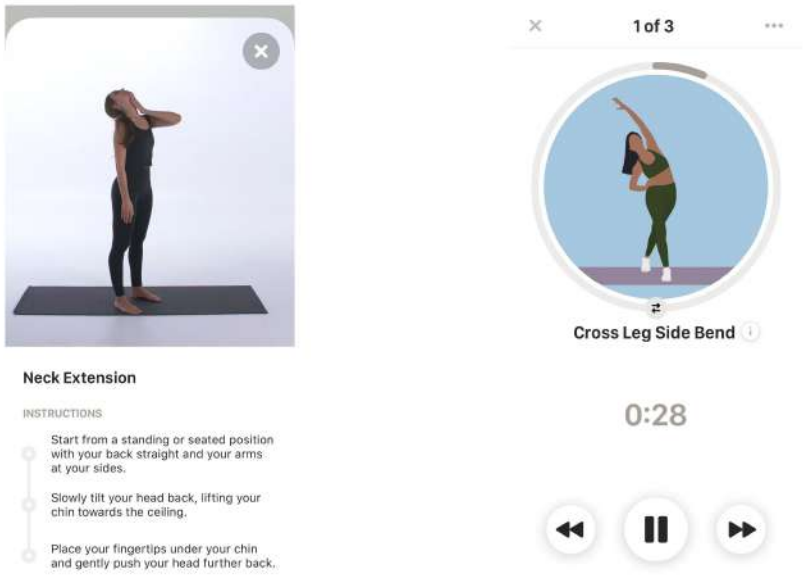


Fig. 11. Screenshot for the bend application. (left) instruction page. (right) The page displayed in the stretching session.

Appendix E: Screenshot of Bend Baseline

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